

The Language of Business Analytics

Research Assignment 1

Foundational Concepts of Analytics

1. **Business Analytics**- It's a tool or a technique that businesses use to analyse their data to perform the business performance and assist in making strategic decision-making. And help data professionals on how to interpret and analyse the information.

Example: Business Analytics in the insurance industry

- Insurance companies use predictive analytics to predict the likelihood of a claim. And the insurer also uses it for Underwriting and pricing decisions by doing risk assessments on clients whether to accept or reject the policies

Source: [Business Analytics: What It Is & Why It's Important](#) (By Matt Gavin on July 16, 2019)

2. **Data analytics**- Its process of collecting, transforming and organizing raw data into useful information that you can use to make business decisions and meaningful insights.

Source: [What is Data Analytics - GeeksforGeeks](#) 25 August 2026

Example: Data Analytics in the Hotel Industry

- Hotels use data analytics to better understand customer behaviour and improve the guest experience. By analysing customer reviews and booking history, hotels can identify guest preferences, common concerns, and opportunities to improve service delivery. They can also track cancellation patterns to understand why customers cancel bookings, helping them introduce strategies to retain clients and improve overall service quality.

Source: [30+ Real-World Data Analytics Case Studies & Business Analytics Examples \[2026\]](#)

3. **Business Intelligence**- Its a technology that can collect and analyze business data. It transforms the raw data into data insights that can be used to identify trends and solve problems. It uses AI and machine learning to do automation analysis

Source: [What is Business Intelligence? A Complete Guide | Qlik](#)

Example: Business Intelligence in the Production company

- **Reporting improvement:** The production company was experiencing challenges with its manual reporting process, which was time-consuming and inefficient. To improve efficiency, the company replaced manual reporting with an automated Business Intelligence system. This reduced the time spent processing reports and allowed the business to access information more quickly for decision-making.

Source: [Business Intelligence \(BI\): Tools, Types, Benefits, and Applications](#)

4. **Data-driven Decision Making** - It is a business process that uses data analysis and insights to help businesses make informed decisions. Instead of relying only on past experience or assumptions, organizations use data analysis to reduce risks, improve accuracy, and make better predictions. Its process doesn't depend only on Data analysis but also on Data quality and governance.

Example: Amazon

- **Amazon collect and analyse customer data to make decisions related to their product recommendations for better pricing and improvements on their products**

Source: [What Is Data-Driven Decision-Making?](#)

5. **Big Data-** it's a complex and large data set that is structured and unstructured data and cannot be contained in traditional data management systems.

Examples: Government departments

- **Government departments can use large datasets collected from communities to better understand public needs and improve service delivery. For example, traffic data can help identify roads that require maintenance, highlight safety concerns, and support better planning for safer and more efficient transport systems.**

Source: [What Is Big Data?](#)

6. **Descriptive Analytics** - Descriptive analytics is a statistical approach that uses historical and current data to identify patterns, trends, and changes within a business. It helps management understand what has happened in the organization and supports more informed decision-making. **Mostly it ask question like "What Happened"**

Example: Financial performance

Financials statements are historical reports that have financial information of the organization that they can use to check the financial stability or health of the organization.

Source: [Descriptive Analytics: What It Is and Related Terms](#)

- 7. Diagnostic Analytics-** It's a data analysis process that identifies why events and outcomes have happened. It discovers the root causes of the events and discovers patterns and explains the relationship between the data sets. **It answers "Why did this happen"**

Examples: Improving the culture of the organization

- The organization can conduct anonymous surveys to understand employees' skills needs, views on company benefits, and concerns raised during exit interviews. This information can help identify the factors contributing to employee turnover and support improvements in the organization's culture.

Source: [What Is Diagnostic Analytics? 4 Examples | HBS Online](#)

- 8. Predictive Analytics-** It's a data analysis process that uses data to predict for future outcomes and trends; it uses historical data for forecasting what will happen. It can rely on Data analytics, AI, and machine learning to calculate the future outcomes. **It answers "What will happen"**

Example: Insurance

- **Insurance companies use predictive analytics to estimate which customers are likely to make a claim by analysing past claims and customer information. This helps them with pricing policies and manage risk.**

Source: [What Is Predictive Analytics? Benefits, Examples, and More | Coursera](#)

- 9. Prescriptive Analytics-** It's an analysis tool where predictions are used to recommend the next things to be done. It's a combination of descriptive, diagnostic and predictive analytics. **It answers, "What should we do next?"**

Example: Banking

- Its can be used to flag fraud, based on all the transactions that a customer makes in a month, they use the transactional data to analyses the patterns of how the customer uses his/her credit card. That will mitigate the risks by identify any anomalies or trends.

Source: [What Is Prescriptive Analytics? 6 Examples | HBS Online](#)

- 10. Data Literacy-** Is the ability to Understand, work, analyze, visualize and apply data. It's not a technical skill but it's a necessary tool that enables one to transform raw data into meaningful insights.

Example: AIRBNB

- Airbnb didn't have any data literacy for their users. They have launched an in-house Data literacy initiative called Data University. This initiative has boosted their skills in understanding, interpreting and using their data

Source: [Data literacy - Wikipedia](#)

11. Structured Data- It's a data that is organized, clear and predefined format. Its data that easy to read by the software and humans. It's a format that comes in rows and columns.

Example: Healthcare Records

- Medical systems store patient data in structured formats like treatment plans. This plan sits in a defined field for tracking and comparison.

Source: [What is Structured Data? - Structured Data Explained - AWS](#)

12. Unstructured Data- It's a dataset that is stored in tabular formats, like Excel spreadsheet and SQL Database. It doesn't have pre-defined data model or organized, and very heavy contains data like Dates, numbers.

Example: Healthcare Record

- Companies uses predictive analysis to do their future forecast to identify the risks using their historical data. They use unstructured data to analyses how specific diseases can be treated

Source: [Unstructured data - Wikipedia](#)

13. Data Warehouse- Its central place where an organization keeps all its organized data, it gathers all the information from different departments and stores it in one location.

Examples: Healthcare

- Healthcare warehouses combine health records, claims and lab records to make an analysis high risk patients for preventive care and to analyze treatments to improve chronic diseases.

Source: [Data Warehouse: Definition, Uses, and Examples | Coursera](#)

14. Data Mining-It's a technique that uses algorithms and statistical analysis to analyse large data sets to uncover patterns, anomalies and other insights. It's mostly used by organizations to make predictions.

Examples: Financial Institutions

- **Data mining is used to manage risks and protect against fraud. Analyze transaction data to identify unusual behavior that may indicate fraudulent activities and uses predictive models to support credit risk assessment.**

Source: [Data Mining: What Is It and How Does It Work?](#)

- 15. Data Visualisation- It converts raw data into visuals like graphs, dashboard and reveals patterns, outliers and relationships that you might not see on raw data. Its simplify complex data and makes it clear to communicate the data insights.**

Examples: Sports

- **Help the coaches and sport analysts to analyse the players' statistics and team performance.**

Source: [Introduction to Data Visualization - GeeksforGeeks](#)

- 16. Dashboard- It displays various types of visual data in one place and presents key performance indicators that show how a business is performing.**

Source: [Dashboard \(computing\) - Wikipedia](#)

- 17. Metric- A quantifiable measure used to assess system performance. Measures the success of everyday business activities that support the KPIs**

Examples: Insurance

- **Insurers track metrics such as claim frequency and loss ratios to monitor performance and manage risk.**

Source: [What is a Metric in Analytics ? - GeeksforGeeks](#)

- 18. Key Performance Indicator- it is a quantifiable measure of performance over time for specific objectives and success on organization activities.**

Examples: Process

- **It measures the cycle time- Measures how long it takes to complete a task from start to finish.**

Source: [What Are KPIs? Defining Key Performance Indicators \[2026\] • Asana](#)

- 19. Benchmark- It is process that companies use to compare their performance against other companies that operates in the same line of business and having the same customers.**

Examples: Product Benchmarking

- The companies use competitors' offers that have the same products in the same industry in order for them to upgrade their current product to be on the same level with the competitor.

Source: [What Is Benchmarking? Types, Benefits, and Practical Use Cases](#)

20. Segmentation- It's a process of dividing large datasets into smaller groups.

Examples: Psychographic Segmentation

- Nike produces a wide range of athletic gear and footwear for various lifestyles. They produce shoes for athletes, for fashion wear.

Source: [What Is Segmentation? Definition, Types, and Uses - ScienceInsights](#)

21. Conversion Rate- is the percentage of users who complete a desired action, such as making a purchase, signing up, or requesting a quote.

Examples: Insurance

- If 50 out of 1,000 website visitors request an insurance quote, the conversion rate is 5%.

Source: [What Is Conversion Rate? | Coursera](#)

22. Customer Acquisition Cost (CAC)- It is a cost that is related to acquiring new customers including all the expenses incurred to acquire all these new customers.

$$CAC = \text{Cost of sales} + \text{Cost of marketing} / \text{Number of new customers}$$

Example

- Avon is a company that sells cosmetics. Between January-March 2026, they ran several new campaigns, spending R40,000 on sales and R20,000 on marketing their products. During that period, they earned 300 new customers. Thus, according to the customer acquisition cost formula,
CAC of Rose Valley during that period = $(R40,000 + R20,000) / 300$
= $R60,000/300$
=R200

Source: [Customer Acquisition Cost: What Is It, Formula, 2 Examples](#)

23. Customer Lifetime Value (CLV / LTV)- It's a revenue a customer is expected to spend in the business over time. It also assist to identify which customers will drive the profits and the churn risks.

Example

- The company wants to target young adults between 18—25 years. Their research is to check if they will stay for (18 months) on average with a certain price.

Source: [Customer lifetime value \(CLV\): What it is + how to calculate it](#)

24. Churn Rate- Is the rate on which customers cancel to do their business with the organization. The churn rate of a company should be zero.

Example

- If an insurance company has 200 customers at the beginning of the month and 10 customers cancel their policies during that month, the churn rate is 5% ($10 \div 200 \times 100$).

Source: [Mastering Churn Rate: Definitions and Calculation Techniques](#)

25. Customer Retention Rate- Is the rate of customers that a company keeps over a specific period. It assists the organization to know how their customers are staying in the company. It's a number of customers at the beginning of a specific period to the end of that period not including new customers.

Example

- Company starts the year with 1000 customers and loses 200 customers during the year, then the retention rate for the year is 80%.

Source: [How To Calculate & Improve Your Customer Retention Rate – Forbes Advisor](#)

26. Net Promoter Score (NPS)- It's a metric that is used in customer experience programs. It's a measurement used to ask customers how likely they will recommend the products and services.

Example

- Furniture company used NPS survey to identify unhappy customers to address their concerns, they used mobile surveys via SMS to resolve the issues and that improved their NPS from 25 to 70.

Source: [What is Net Promoter Score \(NPS\)? How to Calculate & Interpret](#)

27. Average Order Value (AOV)- It's an e-commerce metric that calculates the amount of money customers spend per transactions when purchasing from an online store.

Example

- Woolworths online store operates for a month, during the period they did R300 000 of 200 orders.

$$\begin{aligned}\text{AOV} &= \text{Total revenue} / \text{Total numbers of orders} \\ &= 300\,000 / 200 \\ &= 1500\end{aligned}$$

Source: [Average Order Value \(AOV\) - Definition, Formula, How To Increase](#)

28. Sales Funnel- It is the steps a customer follows from first learning about a product or service until they decide to buy it.

Example

- A customer sees an advert for a new cellphone, visits the store's website, compares the available phones, and then buys one. These steps show the customer moving through the sales funnel.

Source: [What is a Sales Funnel? Complete Guide + Examples | Salesforce](#)

29. Marketing Campaign- It's a process that a product or service uses to promote a brand. It helps the organization to achieve marketing goals of the business.

Example

- Coca-Cola(Share a Coke) designed personalized bottles with individual names, that encouraged the customers to purchase and share the brands and post their pictures on social media. That increased their sales.

Source: [Marketing Campaign | Meaning, Types, Components & Benefits](#)

30. Post-Campaign Analysis- It is the process of evaluating campaign data and results to measure performance, identify areas for improvement, and make better decisions for future campaigns.

Example

- After an insurance company runs a social media campaign promoting vehicle insurance, it compares the campaign's advertising cost, number of quote requests, policies sold, and conversion rate. The analysis shows which advertisements performed best and helps the company improve the targeting and budget of future campaigns.

Source: [Post-Campaign Analysis: Data to Insights - growth-onomics](#)

31. Incremental Revenue- It is the extra income a business earns after doing a new activity, such as running a marketing campaign, launching a new product, or changing prices.

Example

- An insurance company normally earns R100 000 per month. After running a marketing campaign, it earns R130 000. The incremental revenue is R30 000 (R130 000 - R100 000).

Source: [What Is Incremental Revenue? A Complete Guide for Businesses – Money Shark](#)

32. Return on Investment (ROI)- It is a measure used to check how much profit or loss a business makes from the money it invested.

Example

- An insurance company spends R20 000 on a marketing campaign and earns R30 000 from new policies. The profit is R10 000, so the ROI is 50% ($R10\,000 \div R20\,000 \times 100$).

Source: [What Is Return on Investment \(ROI\) and How to Calculate It](#)

33. Return on Ad Spend (ROAS)- It is a measure used to check how much money a business earns from the money spent on advertising.

Example

- An insurance company spends R10 000 on online advertising and earns R40 000 from policies sold through the adverts. The ROAS is 4 ($R40\,000 \div R10\,000$), meaning the company earns R4 for every R1 spent on advertising.

Source: [Return on Ad Spend \(ROAS\): Formula, Benchmarks, and How to Improve It \[2026 Guide\] - TLinky](#)

34. Click-Through Rate (CTR)- It is the percentage of people who click on an advert or link after seeing it. It helps a business check if the message attracts customers.

Example

- An insurance company sends an online advert to 1 000 people, and 50 people click on the link. The CTR is 5% ($50 \div 1\,000 \times 100$).

Source: [Click-Through Rate \(CTR\) | MetricHQ](#)

35. Bounce Rate- It is the percentage of people who visit a website and leave without clicking on anything or going to another page.

Example

- An insurance website receives 1 000 visitors, and 400 visitors leave without clicking on another page. The bounce rate is 40% ($400 \div 1\,000 \times 100$).

Source: [What is Bounce Rate? \(+ Example & Benchmark\)](#)

36. A/B Testing- It is a method of comparing two versions of something to see which one performs better.

Example

- An online clothing store tests two versions of a sale advert. Advert A uses a red button, while Advert B uses a blue button. More customers click the blue button, so the store uses Advert B.

Source: <https://www.geeksforgeeks.org/blogs/what-is-a-b-testing/>

37. Lead Generation- It is the process of attracting people who may be interested in a business's products or services and collecting their details so the business can contact them.

Example

- A clothing store offers customers a 10% discount when they enter their name and email address on its website. The store can then contact these potential customers about new products and special offers.

Source: [Lead Generation: Meaning, Examples, and How to Get Started | Coursera](#)

38. Marketing Attribution- It is a method used to identify which marketing channel or advert helped a customer decide to buy a product or service.

Example

- A restaurant advertises a meal special on radio, social media, and through SMS. Most customers use the discount code sent by SMS when placing their orders, so the restaurant knows that the SMS campaign brought in the most sales.

Source: [What is Marketing Attribution? A Complete Guide – Blog](#)